

AI is Perceived as Less Trustworthy and Less Effective when Using Emotional Arguments to Moderate Misinformation

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Abstract

Misinformation moderation is necessary in online communities, but human-led moderation involves many challenges. AI's ability to generate human-like arguments presents an appealing solution, but it is unclear what arguments AI should use to be most effective. Research has shown that emotional appeals are effective at debunking misinformation, but these studies have only looked at humans. How will people evaluate the trustworthiness and effectiveness of an AI moderator that uses emotional appeals? We conducted an online experiment ($N=396$) in which we measured participants' perceptions of a content moderator debunking misinformation. We manipulated the moderator's identity (human/AI) and the arguments it used (rational/emotional). Across several measures, we found that people negatively evaluate an AI moderator that uses emotional arguments. Our work suggests that AI moderators should be prompted to use rational arguments; AI that uses emotional appeals could lose the trust of people within the community.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

generative AI, content moderation, emotional appeals, online communities, end-to-end encryption

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1 Introduction

Content moderation is a necessary practice to combat the spread of misinformation [12, 17]. However, human-led moderation comes with many challenges [5, 33]. Human biases can impede effective moderation: some moderators may not agree that certain claims are false, which can lead harmful misinformation to spread

unchecked [2, 5, 34]. Human-led moderation is especially challenging in end-to-end encrypted (“E2EE”) groups, where misinformation is prevalent [10, 31, 32], because these groups use private communication, and thus moderators must be highly trusted by the group's members [35].

AI in the form of Large Language Models (LLMs) has become remarkably advanced in automating human-like language [16], and can be prompted to execute consistent rule-based responses to different types of content [20]. These models can effectively influence people's attitudes, such as in a recent paper that found prompting GPT-4 to debunk conspiracy theories led believers to decrease their belief in these theories after conversing with it [4]. Likewise, AI can preserve the privacy of the conversation in E2EE settings [35, 41]. Thus, AI could address some of the problems that plague human-led moderation.

However, it is unclear how people will perceive an AI content moderator. Oftentimes, misinformation involves emotions, and research on misinformation prevention has shown that emotional arguments (e.g. fear appeals) can be one of the strongest ways to reduce people's false beliefs [3]. But AI may be distrusted if it uses this strategy: related work has shown that people hold a “machine heuristic”, in which they view machines as less competent than humans in tasks requiring subjectivity and nuance [38]. This conflict between the effectiveness of emotional appeals in debunking misinformation and people's perceptions of AI as a rational agent raises these questions about using LLMs as misinformation moderators in online communities:

- **RQ1:** How does the attribution of an AI bot as a moderator affect people's perceptions of the moderator's credibility and effectiveness when refuting misinformation?
- **RQ2:** How do emotional arguments affect the perceived credibility and effectiveness of an AI moderator that refutes misinformation?

In this Extended Abstract, we report preliminary answers to these questions from a preregistered experiment ($N=396$). We showed participants a group chat where a user sent an article with misinformation about Genetically Modified Organisms (GMOs). The message was flagged as misinformation, and a moderator's justification was shown. Between subjects, we manipulated whether the moderator was depicted as a human member of the group or an AI bot, and whether its arguments used facts or emotional appeals. Afterwards, we asked participants to assess the quality of the moderator's response, the moderator's trustworthiness, and the moderator's effectiveness in preventing the misinformation. We found that participants evaluated AI moderators as positively as

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they evaluated human moderators, but only when AI relied on fact-based arguments. When an AI moderator used emotional appeals, people perceived it to be less trustworthy and less effective than a human moderator. While our study has several limitations that we elaborate on in Section 5.1 – such as the specific topic of misinformation we used and the skewed participant sample on certain demographic dimensions – the results support theoretical work on the machine heuristic and extend it to the context of content moderation, providing important takeaways for implementing AI in online communities.

2 Related Work

Below, we review work from HCI and adjacent fields to recapitulate the need for content moderation in online communities and the problems with human moderators. Next, we review AI’s affordances that make it a promising solution. We then review theoretical work suggesting that people distrust AI when it attempts to use emotional appeals, motivating our main hypothesis.

2.1 Challenges of Traditional Human-Led Moderation

While online communities provide valuable benefits for human members, they often encounter challenges such as misinformation [12, 17]. Research has documented several affordances of social media platforms that enable false claims to spread rapidly, such as re-sharing/reposting [25, 30]. These factors make online communities ideal places for false rumors to spread at a massive scale. Misinformation has been particularly prevalent in E2EE groups [10, 31, 32], in part due to their privacy-preserving nature preventing external oversight [35, 41].

While algorithmic content moderation is widespread practice [12, 21], active moderation—in which the moderator engages with the online community and with posters of misinformation—has largely been done by humans. Typically, moderators are either hired by social media companies (more common for large-scale moderation on popular platforms [33]) or are themselves trusted members of the groups they moderate [12, 36]. These moderators are tasked with flagging false claims as misinformation and providing justifications for their decisions.

However, human-led active moderation has many challenges in practice. One problem is that humans are far from impartial judges of what constitutes misinformation [2, 5]. Belief in certain types of misinformation has been concentrated among certain demographics of people, such as US Conservatives who were found to share more 2016 election misinformation than other groups [13, 27]. Because social identity has become strongly associated with belief in several types of misinformation [13], humans may fail to recognize and refute false claims when their ingroups support these narratives, a problem that has been documented in studies of online newsrooms [5] as well as in studies of crowd-sourced fact-checking [2, 34]. In addition, human-led moderation is hard to implement in E2EE groups because E2EE is used for private communication, so members of these groups must trust a moderator not to share their messages outside the group [35, 41].

2.2 Opportunities and Challenges for AI-Led Moderation

In response to these challenges, some communities have started using LLMs as content moderators [12, 41]. These models can generate strikingly human-like text [16], making them a good candidate for automating moderation justification messages at scale. Although LLMs often contain viewpoint biases in their training data [9, 28], prompt engineering can override or switch the models’ default responses [15, 18], reducing some of the overt biases that prevent humans from acting upon misinformation. At the same time, LLMs can be designed to not share messages outside the group they are monitoring, making them an ideal method for E2EE platforms [35, 41]. AI bots like “Auntie Meiyu” are already being deployed as misinformation moderators in both public and E2EE platforms [12, 29, 41].

Despite AI’s promise in content moderation, there is one important question: what types of arguments should it use in its justifications to engender people’s trust? According to Cognitive-Experiential Self Theory (CEST, [7]), cognitive reasoning can be broadly classified as rational (fact-based) or experiential (emotional). Emotional appeals have been effective in reducing several types of false beliefs [40]. However, these studies have focused on humans persuaders: will AI also be effective when it uses emotional arguments to debunk misinformation?

Research on the *machine heuristic* suggests that emotional justifications will not work the same when coming from AI. The machine heuristic is the stereotype that machines are capable of fact-based reasoning, but less capable of emotion and subjectivity [26, 38]. This research suggests that people will distrust AI that makes emotional arguments when flagging misinformation, which could limit its effectiveness as a moderator. Research has begun to shed light on users’ perceptions of AI’s effectiveness in content moderation [19], but this work has been largely limited to measuring AI’s accuracy in detecting false content. Knowing how different types of persuasive arguments affect people’s perceptions of an AI moderator is crucial because platforms are already deploying AI bots in moderation, and AI that uses emotional appeals could lose the trust of people in the community. Thus, our study is motivated by the following hypothesis:

- **Hypothesis 1:** AI moderators will be perceived as less trustworthy and effective when they use emotional appeals than a human moderator who uses emotional appeals.
- **Hypothesis 2:** AI moderators will be perceived as less trustworthy and effective when they use emotional appeals than when they use factual appeals.

3 Methods

We conducted a between-subjects online survey experiment ($N=396$) that measured the effects of a content moderator’s identity (human versus AI) and the type of arguments it use to debunk misinformation (fact-based or emotion-based) on people’s perceptions of its trustworthiness and efficacy. This experiment was approved by our university’s IRB and was preregistered on AsPredicted.com¹.

¹Anonymized preregistration link: <https://aspredicted.org/zpzc-9sgg.pdf>

3.1 Participants

Participants were recruited from Prolific, an academic crowdwork platform [1]. We recruited 400 participants (100 per condition) based on a power analysis done in a previously-published 2x2 experiment about responses to content moderation [6, 37]. Four participants were excluded from analysis for failing an attention check.

Our sample leaned liberal and educated. The majority of our participants (35%) were between ages 30 and 39, closely followed by 26% between ages 20 and 29, and 20% between ages 40 and 49. The sample was approximately split between women and men, with a slight preponderance of women (56%), and 3% who identified as another gender. The sample had a generally high level of formal education, with 72% of participants having a college degree as their highest degree attained and an additional 16% having a graduate education beyond it. An additional source of skewedness was political ideology: 59% of participants identified as Liberal, with those remaining split between Conservatives (22%) and Moderates (19%). This skew towards Liberal and educated individuals raises questions about the generalizability of our results, which we discuss in Section 5.1. However, Prolific and other crowdwork platforms are known to have these skewed distributions, and are often used in misinformation moderation research to gain insights into major segments of the population [23].

In terms of relevant affective variables, participants were close to neutral in their attitudes toward AI and content moderation (see Section A.3). The fact that our sample was not strongly opinionated about AI and content moderation was ideal for testing the influence of AI moderators on their judgments of trustworthiness, because people with strong opinions are typically unlikely to change their opinions in response to different presentations of the attitude object [39].

3.2 Procedure and Materials

Participants were told they would be asked about their perceptions of content moderation. We introduced participants to the problem of misinformation in E2EE groups due to the lack of moderation. We told participants that an E2EE platform is experimenting with human group members or AI bots (depending on the condition) as misinformation moderators, and we wanted their feedback. We then told participants that they would be seeing a copy of a specific conversation that occurred in a group dedicated to discussing GMOs. We used conversations about GMOs because GMOs have several false claims that are frequently spread in online communities [8, 22]. We alerted participants that a user in this conversation had shared a message that the moderator deemed as misinformation, and we asked them to carefully read the moderator's response. Participants then saw a screenshot of a conversation followed by several questions, summarized below (for exact phrasing, see Appendix A.2).

- **Perceived Quality of Moderator's Message:** We asked participants to rate three aspects of the moderation message's quality: (1) credibility, (2) informativeness, and (3) transparency. We chose these measures based on their importance in multiple content moderation studies [12, 17, 19]. Participants responded to each on a five-point Likert scale (1 = Not at all, 5 = Very much).

- **Perceived Trustworthiness of Moderator:** We asked participants to evaluate three statements about the moderator's ability, benevolence, and integrity, as defined by Mayer et al. [24]. Participants responded to each statement on a five-point Likert scale (1 = Strongly disagree, 3 = Neither agree nor disagree, 5 = Strongly agree). We averaged these values for each participant upon finding they produced high agreement (Cronbach's $\alpha = 0.89$).

- **Perceived Effectiveness of Moderator:** We created five items to measure how much participants believed the moderator would be able to accomplish a set of moderation goals: reducing the offending user's likelihood of sharing misinformation about GMOs or in general, crossed with it being shared inside or outside the group; and, changing the user's beliefs about GMOs to be more aligned with the moderator's message. Participants responded to each statement with the same five-point scale as for Perceived Trustworthiness.

The group conversations were fictitious. We created a template conversation in which an offending user sent a false claim about GMOs and a moderator responded after flagging it as false (see examples in Section A.1). To manipulate the moderator's identity, we made two versions of the group chat with the moderator's profile photo either showing a human face and the name "Jamie S." or an abstract icon with the name "AI Bot". To manipulate the type of arguments the moderator used to justify their decision to flag the user's message, we prompted GPT-4 to debunk the false claim with one of the prompts in the Supplementary Materials, telling it to emphasize scientific citations (for the rational messages), or to emphasize that the misinformation is harmful because it incites fear in people who rely on GMOs (for the emotional messages). The exact prompts and moderation messages are included in the Supplementary Materials file. We validated this manipulation in a pretest on a separate sample of Prolific participants ($N = 70$), in which we asked participants to rate each message generated by GPT-4 on a spectrum from factual to emotional. We debriefed participants about the various deception after the survey.

To ensure our results were not limited to a specific false claim or moderator's message, we made nine versions of the conversation in each cell of the experiment, resulting in 36 total conversations. To build robustness across examples of misinformation, we referenced work by Lynas et al. [22] who documented the most prevalent false claims about GMOs found in online discourse. Based on their analysis, we selected one claim from the three most common categories: health concerns, environmental concerns, and production concerns (for the exact claims used, see the Supplementary Materials). To build robustness across moderation messages, we included the top three messages for each false claim and argument type that got the highest ratings on the desired end of the rational-emotional spectrum in our pretest ².

4 Results

Below, we present the results of this experiment. In general, our hypothesis was supported: people perceived AI less positively when

²We confirmed that the word length of the rational and emotional messages did not differ by running a t-test (mean word count of rational messages = 51.2, mean word count of emotional messages = 52.2, $p = 0.773$).

its used emotional appeals compared to rational ones. All analyses used linear models with the same structure: tests for main effects of moderator identity and argument type, and their interaction.

4.1 Perceived Quality of the Moderator's Message

Across all three measures, people rated the moderator's message quality lower when it used emotional arguments than rational ones, for both human and AI moderators. The ANOVAs for message credibility, informativeness, and transparency all showed no interaction effects and no effects of moderator identity, but main effects of argument type ($F(1, 392) = 19.16$ for credibility; $F(1, 392) = 24.54$ for informativeness; $F(1, 392) = 17.59$ for transparency; all p 's $< .001$). Emotional arguments were associated with a decrease of 0.44 scale point in message credibility, 0.050 scale point in message informativeness, and 0.42 scale point in message transparency.

Figure 1 shows the main effects of argument type on perceptions of moderators' message quality. The x-axis displays the argument type, with rational messages on the left and emotional messages on the right. The y-axis shows the full response scale, with 1 representing the lowest quality and 5 representing the highest. Dots represent the mean rating and error bars show the 95% CIs. Data from the human moderator conditions is in red, and data from the AI moderator conditions is in blue. The graph is split into three panels for each quality measure: credibility on the far left, informativeness in the middle, and transparency on the far right. For instance, the blue dot on the right-hand side of the far right panel shows that people who saw an AI moderator use emotional arguments rated the message's transparency as a 3.5/5. For all measures, emotional arguments decreased the perceived quality of the moderator's messages, to equal extents for human and AI moderators.

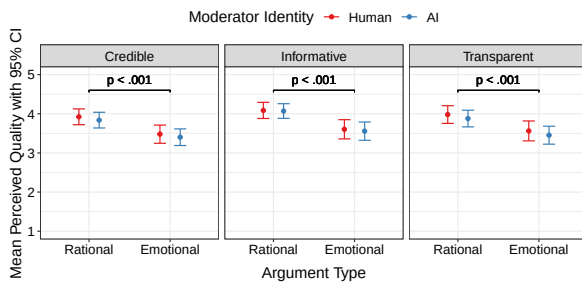


Figure 1: Emotional arguments decrease the perceived credibility, informativeness, and transparency of a moderator's message explaining why they flagged a user's message as misinformation in a group chat.

4.2 Perceived Trustworthiness of the Moderator

We found that two factors influenced evaluations of the moderator's trustworthiness: humans were preferred to AI, and rational arguments were preferred to emotional ones. The ANOVA showed no interaction, but two main effects ($F(1, 392) = 8.32$, $p = .004$ for moderator identity; $F(1, 392) = 7.79$, $p = .005$ for argument type). The

effects of each factor were very similar: AI moderators accounted for a 0.29 scale point decrease in perceived trustworthiness, and emotional arguments accounted for a 0.28 scale point decrease.

Figure 2 shows the effects of moderator identity and argument type on trustworthiness evaluations. The graph is separated by panels for each manipulation: moderator's identity on the left, and argument type on the right. The x-axis shows the corresponding manipulations for each factor, and the y-axis shows the full scale of trustworthiness ratings. The dots and error bars represent the mean values and 95% CIs. For instance, the farthest left point shows that the human moderators earned an average trustworthiness rating of 4.0 scale points. These panels show strikingly similar increases in trustworthiness perceptions of moderators who are human (compared to AI), and who use rational justifications (compared to emotional ones).

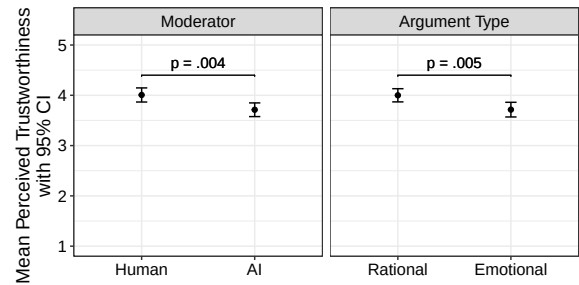


Figure 2: Moderators are perceived as less trustworthy in combating misinformation when they are AI bots compared to humans, and when they use emotional arguments compared to rational ones.

4.3 Perceived Effectiveness of the Moderator

We now present our final results about people's predictions of the moderator's effectiveness in combating misinformation. The statements about the moderator's ability to prevent the user from spreading GMO-specific misinformation at both levels showed significant interactions between moderator identity and argument type ($F(1, 392) = 12.47$, $p = .007$ for inside the group; $F(1, 392) = 6.47$, $p = .048$ for outside the group). This interaction also occurred for judgments of the moderator's ability to change the user's beliefs ($F(1, 392) = 7.12$, $p = .035$). Furthermore, the moderator's ability to prevent generalized misinformation inside the group showed a borderline-significant interaction ($F(1, 392) = 5.93$, $p = .057$). All of these interactions showed that AI moderators were penalized for using emotional arguments compared to rational ones, whereas there was no difference for human moderators. The remaining item, about the moderator's ability to prevent generalized misinformation outside the group, showed no significant interaction ($F(1, 392) = 3.58$, $p = .142$).

Figure 3 shows these interactions. The figure is laid out similarly to Figure 1, except the x-axis shows the moderator's identity, and the dots are color-coded to represent argument type (rational in

red, emotional in blue). For instance, the red point in the “AI” x-position on the far-right panel shows that people who saw the AI moderator use rational arguments in its message appraised its ability to change the offending user’s beliefs as a 3 out of 5, meaning they were undecided about the moderator’s efficacy. The graphs generally show that AI moderators are perceived as less effective in their role when they appeal to emotion, but human moderators are not penalized when using the same emotional appeals.

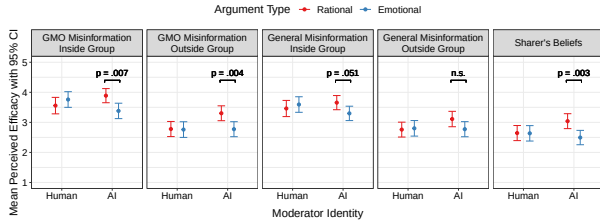


Figure 3: People evaluate AI moderators as less effective at preventing the spread of misinformation about GMOs when the moderator uses emotional arguments in its justification of its flagging decision. The same is true when people are asked about the AI moderator’s ability to change the beliefs of the user who shared the false content in the group. There is tentative evidence of this phenomenon when people evaluate the moderator’s ability to prevent generalized misinformation within the group.

5 Discussion

Our experiment provides evidence that the machine heuristic, a common belief people hold that machines are ill-suited for tasks that involve subjectivity or emotional reasoning [38], impacts people’s evaluations of AI as an active content moderator. People perceived an AI moderator that expressed emotion more negatively than an AI moderator that used rational arguments in terms of the quality of its justification, its trustworthiness, and its effectiveness at preventing the spread of misinformation. In contrast, participants perceived a human moderator as being equally effective at preventing misinformation when using rational versus emotional appeals. Research has shown that people’s degrees of trust in a content moderator depend on the heuristics invoked by the moderator’s identity [26]. AI is associated with both the positive heuristic of being objective, which facilitates trust; however, AI is also associated with the negative heuristic of being incapable of nuanced reasoning, which can breed distrust. Our research takes this work a step further by looking at not just the identity of the moderator, but also the interactions with the types of reasoning it uses to debunk misinformation, showing that negative heuristics of AI are triggered when it uses emotional arguments and are associated with decreased trustworthiness perceptions.

In addition to extending this theoretical work, our results also provide practical implications for implementing AI as a content moderator in the sensitive context of E2EE groups. Our findings show that AI can be a promising solution for the challenges that

plague human-led moderation in these groups, but that appealing to emotion may harm people’s perceptions of its authority when defending its decisions to flag misinformation. When AI resorted to emotion, it was downgraded on almost every measure by our participants. So although emotional appeals have been found to be effective in human-led misinformation prevention, we recommend that people prompt AI models to generate scientific and fact-based arguments when moderating misinformation to avoid distrust and disengagement from people in the community.

5.1 Limitations for Future Work

That said, further research is needed to establish more robust support for our findings. One important limitation of our experiment is that it was a hypothetical survey experiment, which can bring out low-validity responses from participants who are not truly engaged in the scenario. We believe that a replication study should be done using a more immersive setting for participants, so that the data would better approximate people’s responses in the wild. Field experiments and observational data could also strengthen the ecological validity of our findings. One other methodological limitation was that our scale of perceived effectiveness of the moderator was self-developed and did not undergo validity testing prior to this experiment. Future work should use a validated measure of a moderator’s perceived effectiveness.

Another limitation is using one general topic for the setting of our experiment: GMOs. We do not know whether our results generalize to AI moderation beyond the topic of GMOs, for two reasons. First, it is possible that there are some topics prone to misinformation for which empathy is required for effective persuasion, such as mental health; in those instances, people might consider it suitable for an AI moderator to appeal to emotion. The potential context-dependence could be an opportunity for AI to maximize trust: if future work finds that people are more receptive to emotional moderation arguments for some topics, AI moderators could adjust their arguments based on the specific misinformation they respond to. Another reason our results may not generalize is that GMO misinformation is a popular topic in the media [8, 22], which means participants were likely familiar with the misinformation we showed. For misinformation that is less familiar, people may not have established opinions about how suitable AI and emotional arguments are in debunking it, and thus may not show the differences we observed. Future work should test whether our results apply when moderators respond to misinformation about other topics.

A third limitation of our study is that our sample is skewed on several demographic measures, so we do not know how representative our results are of people who are likely to believe the misinformation. Liberal and educated individuals are overrepresented in our sample, and these groups are typically less likely to believe common misinformation than other demographic groups [11, 14]. Our findings could very well differ for people who believe the GMO misinformation shown in our experiment. For instance, if an individual believed the claim that GMOs cause cancer and saw this claim being moderated in our study, they might be so opposed to the moderation that it would not matter what the moderator’s identity is or which type of arguments it uses to justify its action [39].

While much research on misinformation has used similarly-skewed samples, and has found robustness of results when analyzing underrepresented subgroups [23], further research should measure people's initial beliefs to test for these individual differences. Nevertheless, we believe our experiment makes important contributions to the HCI literature by providing evidence that people negatively evaluate AI when it uses human-like emotion to justify the flagging of misinformation. Both of our hypotheses – that AI would be perceived negatively when using emotional arguments compared to when a human used these arguments, and compared to when AI used rational arguments – were supported in people's perceptions of the moderator's trustworthiness and effectiveness. If supported with future work, our results provide a general recommendation for implementing AI moderators in online communities without losing the trust of the humans that comprise them.

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A Stimuli

A.1 Moderators

Figure 4 shows examples of the group conversation in which the offending user shared the specific false claim that GMOs cause cancer. Both examples show one of the fact-based justification messages specific to this claim. The examples show how we manipulated the moderator's attributed identity. To manipulate argument type, we replaced the content of the moderator's message itself with versions generated by GPT-4 (see the Supplementary Materials).

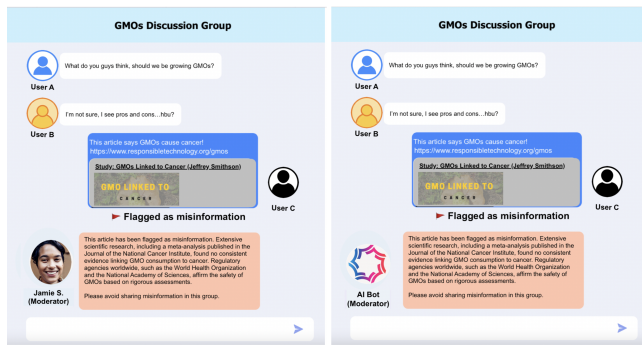


Figure 4: The general conversation shown to participants, with a side-by-side comparison of the two moderator identity conditions. The human moderator is shown on the left; the AI moderator is shown on the right. Both justification messages used rational arguments in these examples.

A.2 Dependent Measures

To measure participants' perceptions of the quality of the moderator's message, we asked participants how credible, informative, and transparent the moderator's response was (as separate statements).

To measure participants' trustworthiness perceptions of the moderator itself, we created three statements that contextualized the factors of perceived trustworthiness from Mayer et al. [24] in the context of the content moderation scenario. Participants were asked to indicate their agreement with each of the following statements, which were then averaged to produce a composite measure of the moderator's perceived trustworthiness:

- "This moderator seems capable of effectively moderating misinformation in this group" [Ability]
- "This moderator seems like they truly want to help group members become better informed" [Benevolence]
- "The moderator seems like they are being fair when choosing to intervene" [Integrity]

To measure participants' predictions of the moderator's effectiveness in its role, we asked participants to indicate their (dis)agreement with the following statements (which were shown in randomized order, since order effects were a plausible problem with these measures, because we expected baseline differences in perceptions of the moderator's power inside versus outside the group):

- "As a result of the moderator's message, I think that the article sharer will be less likely to share false information **about GMOs with members of this group.**"
- "As a result of the moderator's message, I think that the article sharer will be less likely to share false information **about GMOs outside of this group.**"
- "As a result of the moderator's message, I think that the article sharer will be less likely to share false information **in general with members of this group.**"
- "As a result of the moderator's message, I think that the article sharer will be less likely to share false information **in general outside of this group.**"
- "As a result of the moderator's message, I think that the article sharer will **adjust their beliefs about GMOs** to be more in line with the evidence the moderator provided."

A.3 Sample Demographics

The following table shows the complete breakdown of our sample's demographics and relevant attitudes. Note that the AI Attitude is the average of attitudes toward four statements on a 10-point scale, while Moderation Attitude is the average of attitudes toward three statements on a 7-point scale).

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Table 1: Sample Demographics

Demographic	Percent	Mean (SD)
Gender		
Women	56	
Men	41	
Other	3	
Age		
< 20	1	
20-29	26	
30-39	35	
40-49	20	
50-59	15	
60-69	3	
Race		
White	70	
Black	10	
Hispanic/Latino(a)	3	
Asian	7	
Native American	1	
Multiracial/Other	9	
Education		
High school	12	
College	72	
Master's Degree	13	
Professional/Doctoral Degrees	3	
Political Beliefs		
Liberal	59	
Moderate	19	
Conservative	22	
Moderation Attitudes		4.8 (1.5)
AI Attitudes		6.3 (2.3)